

IoT, wearables, driverless cars were all on the horizon back then as well, it would have been pretty difficult to predict Slack being available as a smartwatch app, or a user experiencing problems while streaming a show on Netflix in a high speed train. Similarly, nobody really knows what kind of applications will be necessary in 2030, around the time that we will see 6G moving from research to deployment. The specifications and features of 6G will depend on the problems and shortcomings that we will encounter and see during the deployment of 5G and its use over the next ten years. Emerging technologies can only give us a general idea, and nobody knows for sure why 6G is necessary and what it will look like. As a matter of fact, even the 5G rollout, its adoption and the development of applications needs to be played out, so even experts only know the potential of 5G. 5G is fast, supports a number of devices, and has low latency. It is also being built to be upgradeable and flexible, with future releases. All of this is in theory. In a panel discussion hosted by InterDigital at MWC 2019, experts were asked about the potential role of 6G in the future. Colin Willcock, Head of Radio Network Standardization at Nokia responded, "I don't think that is a relevant technical question. I do think it is a relevant question to say, can we achieve the promise of 5G? And how long will it take? I think, perhaps that is a more interesting question, and there I think the jury is out. The baby has started walking, will it learn to run? I don't know, it might fall over and hurt itself and that is the end of the story. In other words we might get to a point where we have LTE plus 25%, that's all you get with 5G, we have a lot of disappointed users, game over. Wait for 6G." If 5G delivers on everything that it is supposed to, there might not be a need for 6G at all. Right now, we have no idea about the limits of 5G. Still, researchers around the world have already started working towards the next generation network, and there are some solid reasons why "6G" as a concept is already getting relevant. First of all, researchers around the world need to differentiate their efforts towards the next generation of cellular

networks from what is already available. 5G may just turn out to be not open and flexible enough to support emerging applications. Finally, marketers would need a buzzword to sell their products, and we may see a 6G for this reason alone. There are already institutions and labs around the world working towards the next generation of telecom networks.

5G communications use frequency in the range of 600 MHz to 40 GHz. 6G networks would use frequencies in the range of 100 GHz to 1 THz. In the US, the Defense Advanced Research Projects Agency (DARPA) and the Semiconductor Research Corporation (SRC) has set up the ComSenTer lab at UC Berkeley, to investigate terahertz communications. Telecom infrastructure using such high frequencies can support 10 to 1,000 times more wireless devices than 5G networks. At this frequency range, it will be possible to boost up the bitrates to terabits per second. The applications being imagined for the network include a classroom with a 1,000 students simultaneously experiencing an AR lecture, or autonomous vehicles that can detect accidents kilometers down the road despite bad weather conditions. Ali Niknejad, the associate director for ComSenTer says, "our center is simply the next generation of communication and sensing, something that may become 6G".

Ericsson's research division is deeply involved with developing the infrastructure to meet the demands of future applications. One of the areas of focus is mobile edge computing, which requires more computing power and intelligence to be embedded in the massive volumes of IoT devices expected in the future. These devices would need to talk to the cloud less. The eventual destination is a universal computing fabric, where the devices and cloud seamlessly meld together.

The University of Oulu in Finland is among the first in the world to focus on research into 6G technologies. The University also had one of the first 5G test networks in the world. The 6Genesis program is funded for 8 years, and will closely monitor the implementation of 5G, develop the technology

components for 6G, and actively try to accelerate the adoption of 5G as well as 6G. This will involve investigations into the materials and components for the infrastructure and products as well. One of the most important components is the antenna, which started off as a 30 cm long structure in the era of GSM. By the time of 6G, the antenna can be expected to shrink to nanometer scales. There are also certain challenges to overcome, for example terahertz frequency signals are not great at penetrating buildings, and the telecom infrastructure might have to be built into the buildings themselves. Developing the technologies for 6G could have benefits in other areas as well. For example, for most of history radio communications were handled through a single transmitter and a receiver. Since the introduction of WiFi and 3G, there has been an increasing use of multiple-input and multiple-output (MIMO) antennas which simultaneously send and receive more than one data signal on the same channel by firing the data in several directions and taking advantage of the way they bounce off surfaces. At terahertz frequencies, MIMO antennas can be used to create 3D visualisations of the environment, and replace XRay machines at airport security checkpoints, or lead to high resolution medical imaging devices. The program will also be used to investigate distributed intelligent computing and its possible applications. The 6G network is itself expected to become a smart and self-aware nervous system for the world, adapting to different kinds of usage in real time, anticipating future demands, and communicating with engineers the kind of upgrades that it needs. These are just some of the research objectives at Oulu.

The Director of the 6G flagship at the University of Oulu, Matti Latva-aho says, "by 2030s the digital application range explodes and new value chains emerge. Wireless connectivity is driving major societal changes, and the impact of 6G research will be tangible for example in the automation of manufacturing technologies, sustainable energy, e-health, and autonomous transportation – all investigated in the 6G research cluster". 